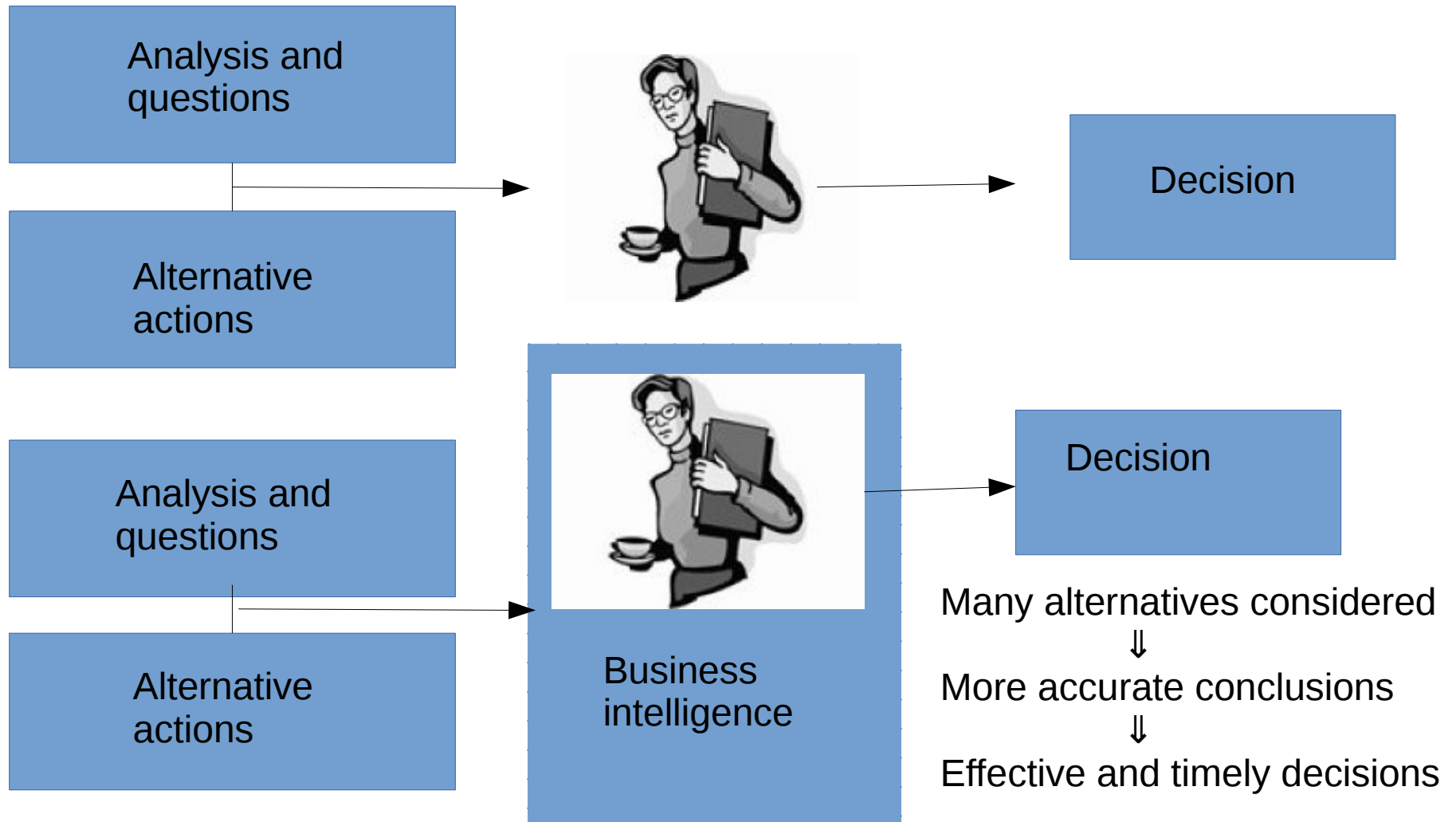


Benefits of a business intelligence system



Data, information and knowledge

- Data. Generally, data represent a structured codification of single primary entities, as well as of transactions involving two or more primary entities.
- Information. Information is the outcome of extraction and processing activities carried out on data, and it appears meaningful for those who receive it in a specific domain.
- Knowledge. Information is transformed into knowledge when it is used to make decisions

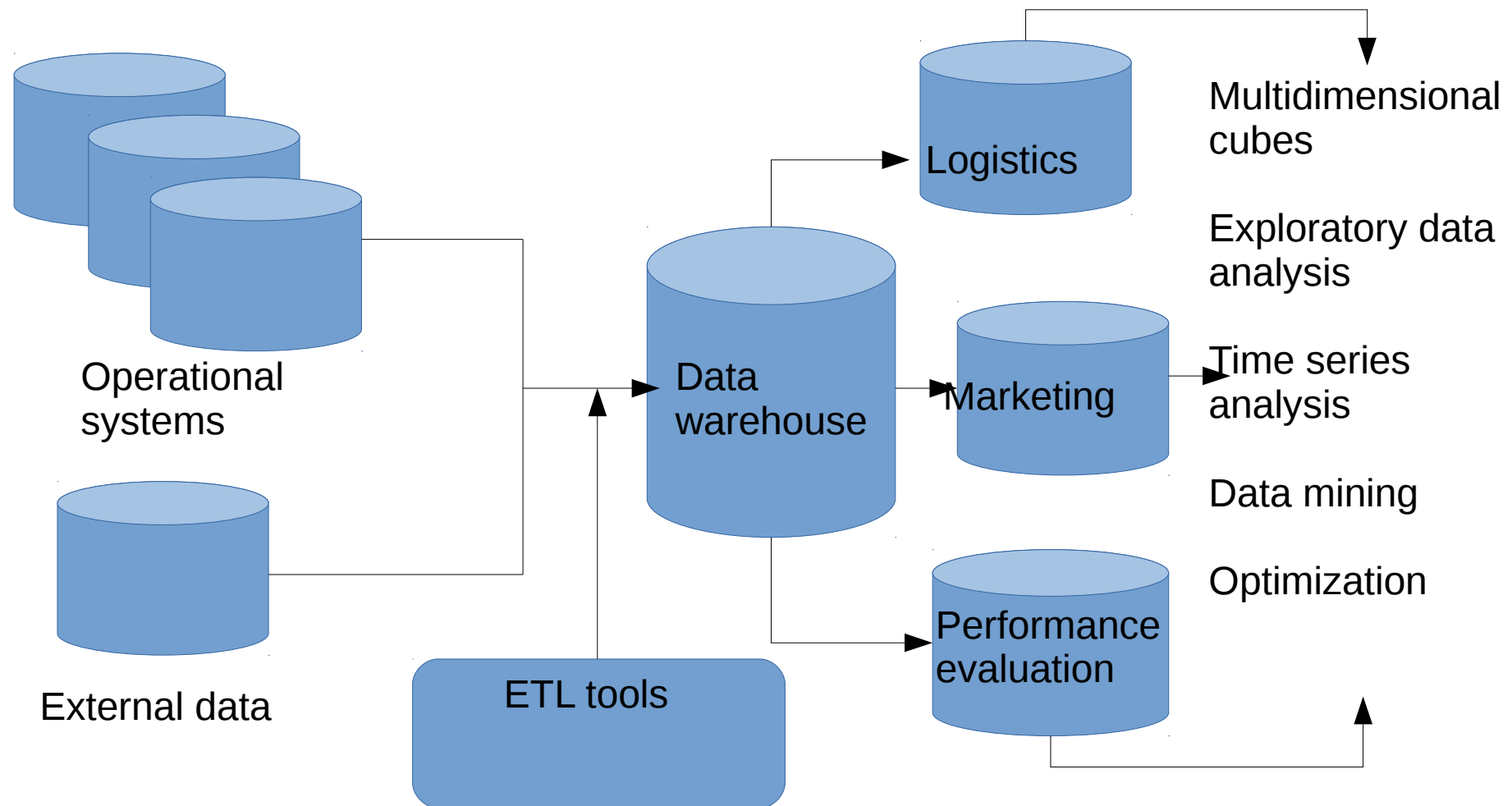
The role of mathematical models

- The rational approach typical of a business intelligence analysis can be summarized schematically in the following main characteristics.
- First, the objectives of the analysis are identified and the performance indicators that will be used to evaluate alternative options are defined.
- Mathematical models are then developed by exploiting the relationships among system control variables, parameters and evaluation metrics.

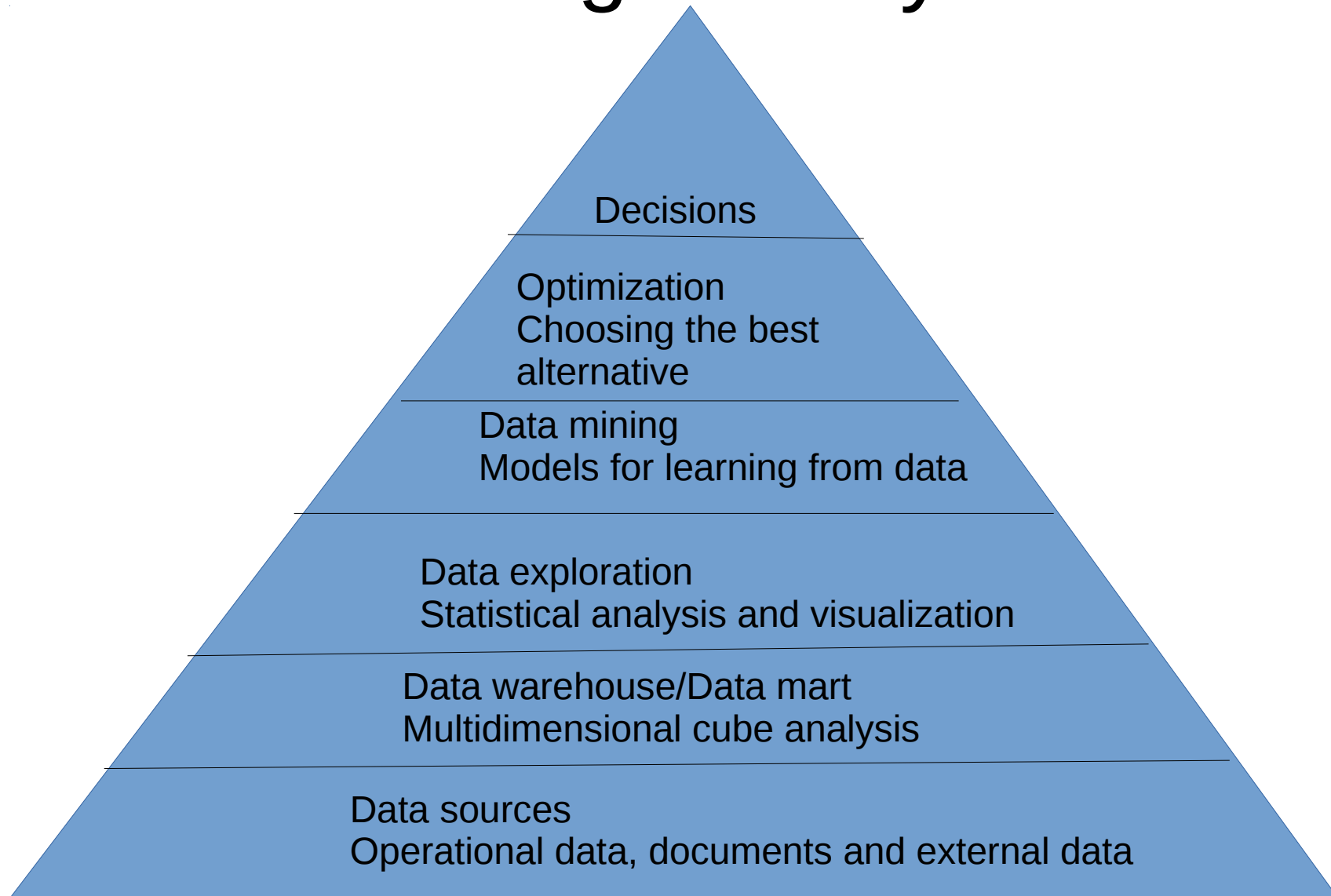
The role of mathematical models

Finally, what-if analyses are carried out to evaluate the effects on the performance determined by variations in the control variables and changes in the parameters.

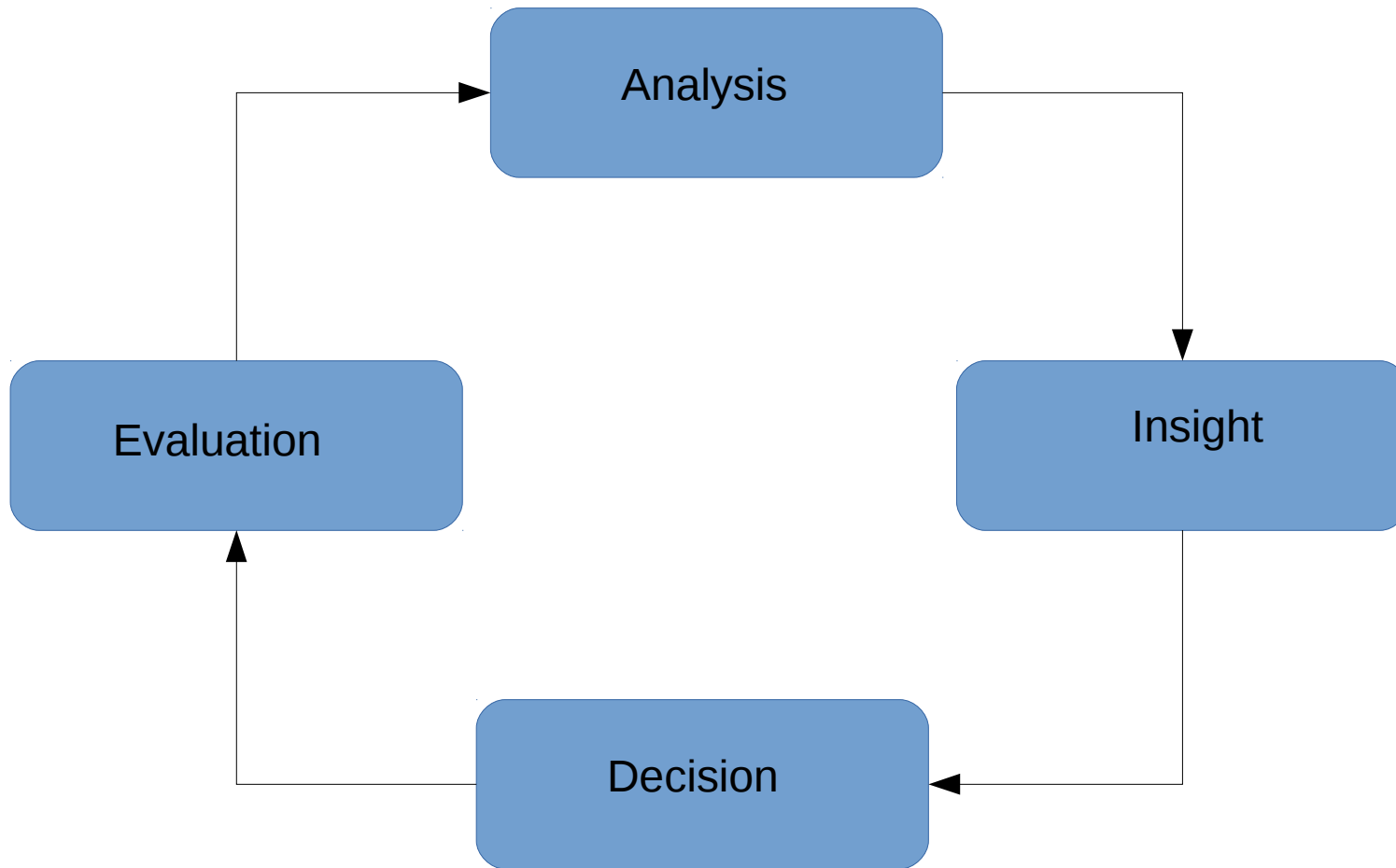
Business intelligence architectures



The main components of a business intelligence system



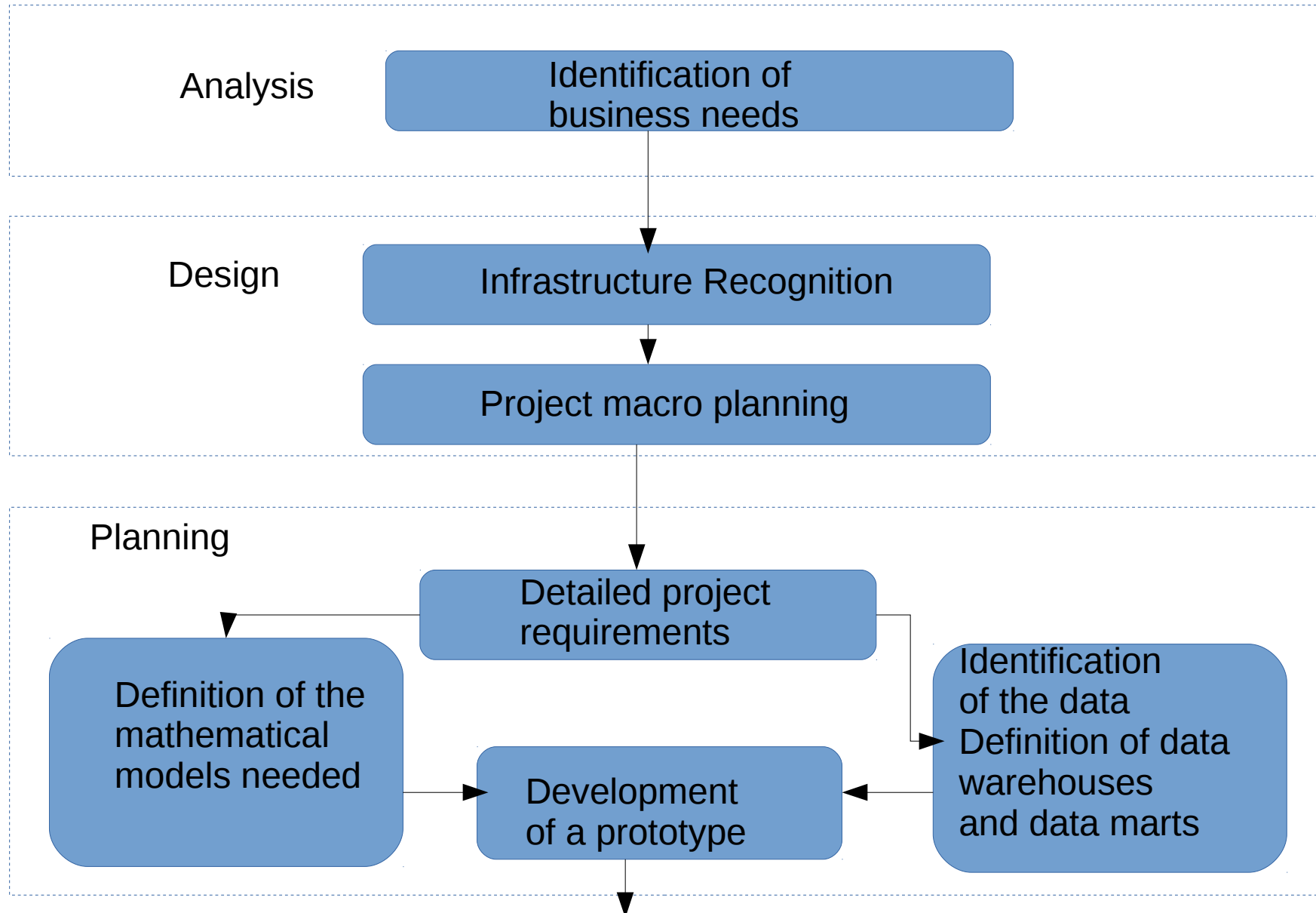
1. Cycle of a business intelligence analysis



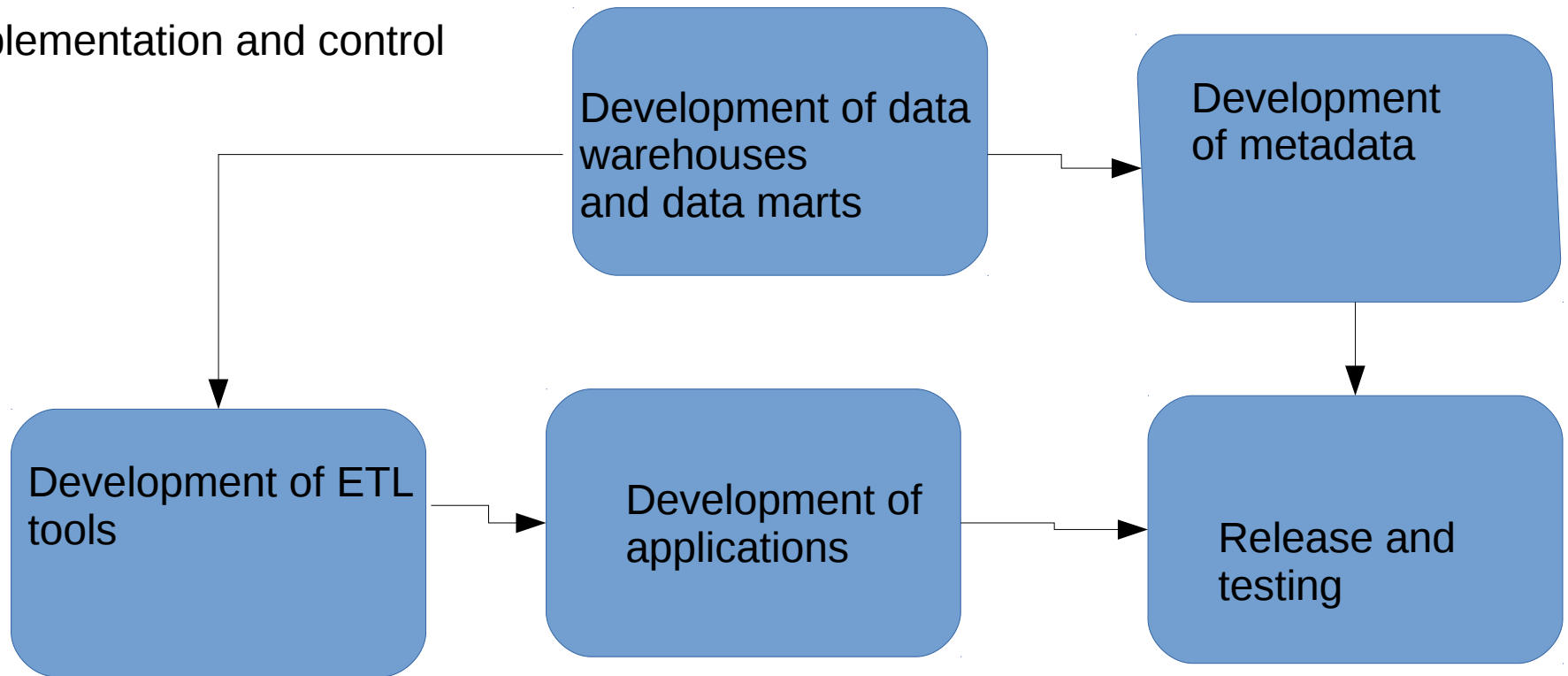
2. Enabling factors in business intelligence projects

- Technologies
- Analytics
- Human resources

Development of a business intelligence system



Implementation and control



Ethics and business intelligence